SUBODH KUMAR SAHU

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Summary

AI/ML-focused Computer Engineering graduate with hands-on experience building LLM-integrated and agentic systems — RAG pipelines using LangChain and LangGraph, and an adaptive LLM-driven decision system for real-time threat classification. Comfortable working through ambiguous, research-oriented problems independently, prototyping quickly, and documenting findings (published IEEE manuscript). Strong Python foundation with backend systems experience (FastAPI, Docker) for turning prototypes into working services.

Education

B.E. in Computer Engineering — **2022-26**
Terna Engineering College, Navi Mumbai — CGPA: 8/10

Higher Secondary Certificate (HSC) — **2022**
Atomic Energy Central School No. 4 — 75.5%

Technical Skills

AI/Agentic Systems:	LLMs, RAG, LangChain, LangGraph, Agent-Driven Decision Systems, PyTorch, Hugging Face Transformers
Data & Analytics:	Pandas, NumPy, Matplotlib, Seaborn, SQL
Core:	Python, Object-Oriented Programming (OOP), Data Structures & Algorithms, REST APIs, Git, SDLC
Backend:	FastAPI, Flask, API Development, Automation Scripting
Databases:	PostgreSQL, MongoDB, ChromaDB
Infra:	Docker, AWS, Linux, GitHub (Actions, CI/CD)
Languages:	C++, C, JavaScript, TypeScript, HTML, CSS

Experience

Software Development Intern — **May 2025 – July 2025**
Bhabha Atomic Research Centre (BARC), Trombay

- Developed **8 modular Python/**C++ software components for a **Bionic Robotic Hand**, improving code modularity and reducing development effort for feature integration by **35%**.
- Built a cross-platform desktop application using **PySide6 (Qt)** with **UART**-based communication, achieving **<80ms** command-response latency for reliable real-time robotic hardware control.
- Tested and debugged **20+** software issues across multiple application modules, reducing runtime failures during validation by **40%**.
- Collaborated with engineers to develop, validate, document, and maintain software components, following structured **SDLC** and version-controlled testing practices.

Projects

Research Paper RAG Assistant — *Python, FastAPI, LangChain, ChromaDB*

- **Built** a document ingestion and retrieval pipeline using **Python** and **FastAPI**, processing **122** research paper chunks with an average semantic search response time of **5.5s**.
- **Designed** a **ChromaDB**-based semantic search system managing **5,000+** vector embeddings, returning relevant results in **350ms**.
- **Orchestrated** multi-step retrieval-and-reasoning workflows using **LangChain**, integrating external LLM APIs (OpenAI, Groq) through reusable REST services.

AI Adaptive Cyber Honeypot — *Python, FastAPI, Docker, PostgreSQL, React, Nginx, Qwen 3.4 7B*

- **Built** an LLM-driven adaptive decision system using **Qwen 3.4 7B** that autonomously classifies malicious payloads and generates real-time response actions without hardcoded rules.
- **Developed** a **FastAPI/PostgreSQL** backend processing **4,200+** simulated security events during testing with **92%** successful service availability.
- **Implemented** REST APIs and structured logging across **12+** API endpoints, reducing debugging time by **30%** through centralized logging and improved observability.

T5 Text Summarizer — *Python, FastAPI, PyTorch, Hugging Face Transformers*

- **Built** a **T5**-based text summarizer reducing document length by **80%** while preserving primary context and key information.
- **Optimized** inference using beam-search decoding, reducing average response time from **3.6s to 2.5s** across CPU, CUDA, and Apple MPS environments.
- **Exposed** the application through **FastAPI** REST APIs, delivering a backend service with a responsive web interface.

Certifications

- **Oracle:** Generative AI Professional | AI Vector Search Professional | Data Science Professional | AI Foundations Associate
- **Google Cloud & AI:** Python AI Assistant | Career Launchpad | Introduction to NLP | Machine Learning Algorithms

Achievements

- **Smart India Hackathon (SIH) 2024 & 2025:** Institute Finalist, developing AI-powered solutions for real-world problem statements in India's largest innovation hackathon.
- **Research Paper (IEEE Manuscript):** Ensemble Learning Approaches for Loan Approval Prediction: A Three-Phase Machine Learning Framework.
- **Winner – Special Mention Award:** Operation Trinetra Hackathon, Anna University (Top Tier out of 170+ teams).
- **E-Cell Member:** Technical and Web Development Team.